

Problem 4.10

You have \$260 in a bank savings account that earns simple interest. You make no subsequent deposits in the account for the next four years, after which you plan to withdraw the entire account balance and buy the latest version of the iPod at a cost of \$299. Find the minimum rate of simple interest that the bank must offer so that you will be sure to have enough money to make the purchase in four years.

Solution

For simple interest,

$$A = P(1 + it).$$

Here,

$$P = 260, \quad A = 299, \quad t = 4.$$

So,

$$299 = 260(1 + 4i).$$

Dividing by 260,

$$\frac{299}{260} = 1 + 4i.$$

Therefore,

$$i = \left(\frac{299}{260} - 1 \right) \div 4.$$

Hence,

$$i = 0.0375.$$

So,

$$i = 3.75\%.$$